

Project

Big data and distributed processing

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Project Themes

- 1. Application that uses distributed database for managing data.
- 2. Application that uses Apache Hadoop for data analysis.



Distributed Database Application

1. Create Application that will use distributed database, for example:
 - Distributed system for cinema reservation.
 - Distributed system for house renting.
 - Distributed system for hospital.
 - Distributed system for library.
 - Distributed system for flight management
- Possible own theme after discussion.

Distributed Database Application - Requirements

- One or two persons
- Console Application (1 person) or GUI (pair).
- At least 2-3 node working in project (real or Docker based).
- Application can work on every node in cluster
- Project must have the following functionalities:
 - Make reservation (make an entry into the database)
 - Update reservation (update an entry into the database)
 - See reservation and see who made it (see an entry in the database)
- (Only for Pairs)
 - Cancellation of reservation/entries
 - Cancellation of more reservations/entries then one at a time.

Distributed Database Application - Requirements

- Error handling and no big delays while processing data.
 - The ability to automatically generate a large load of data (stress test), the created system must handle it.
 - Stress Test 1: The client makes the same request very quickly.
 - Stress Test 2: Two or more clients make the possible requests randomly.
 - Stress Test 3: Immediate occupancy of all seats/reservations on 2 clients.
(It should look as 2 parties/persons try to make as many as possible reservation at a same time and result should be that both are able to make have reservations, not 1 party/person takes all reservations.)
 - Stress Test 4: (only for pairs) constant cancellations and seat occupancy.
 - Stress Test 5: (only for pairs) Make large group cancellation of many reservations.
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Apache Hadoop for data analysis

2. Create **Application** that will use Apache Hadoop

- Find (or create) a large Dataset.
 - Make an analysis, see what's in data and what you can take from it.
 - Look for at least 6 interesting points in data that you want user to browse from.
 - Create an application that user can browse this data. (The accessibility of shown data is part of grade, preferred application that can browse over data.)
- Example: From Car Price dataset you could get:
 - Brands of car and prices
 - Fuel Types and prices
 - Mileage and Brands
 - Number of Doors and Brands
 - Number of previous owners and price
 - Number of sold electrical and oil based cars over the years
- Now application should be able to make user select one of such options and make “queries” to browse this data.
- Recommendation system (not necessary for passing project but will give you higher grade) Have application make 1 recommendation, for example which car is best to buy from this brand.
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Additional Requirements for all Projects

- Description (readme file in git describing in a few sentences what the project does and how to run it).
 - Short report (1-2 pages, description, database schema, problems encountered).
 - Submission by the end of the semester. Late submission will result in a lower grade.
 - Sharing the project with the teacher, preferably in the form of a git repository.
 - Code written by yourself.
 - GitLab: @AdamGodzinski
 - GitHub: @AdamGodzinski
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